

RIYA CHADDHA

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EDUCATION

Master of Science in Information Systems, Northeastern University

May 2025

Bachelor of Engineering in Computer Science, University of Pune

May 2020

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, R, Linux/Bash Scripting, JavaScript
Databases: MySQL, Oracle SQL, SQL Server, NoSQL, Snowflake
Tools & Technologies: Azure Data Factory, Apache Airflow, Talend, Alteryx, Data mining, Data Modeling
Analytics/Visualization/CRM: Power BI, Tableau, Azure Synapse Analytics, Azure Data Lake
Certifications: Machine Learning and Business Intelligence, Microsoft Azure Data Fundamentals

WORK EXPERIENCE

Graduate Teaching Assistant, Northeastern University

Sept 2024 – Present

- Mentored 100+ students in **SQL transaction management** and **data warehousing**, providing guidance, structured assessments, and project feedback using **Azure SQL, Docker, and Snowflake**
- Spearheaded lab sessions and class discussions with the professor, implementing a monthly participation grading system that increased student engagement by 67% and fostering peer learning initiatives to enhance collaboration

Graduate Research Assistant, Northeastern University

May 2024 – Aug 2024

- Optimized ad content classification by developing a **DistilBERT**-based model, achieving 75% annotation agreement on 150+ video ads, enabling **data-driven marketing** decisions
- Built **Tableau** dashboard to visualize ad classification results, highlighting key content themes, and audience engagement patterns (e.g. emotional tone, call-to-action effectiveness) to optimize campaign strategies
- Analyzed 1,000+ survey responses using Tableau to explore potential links between high-stress and shift-based job roles on sleep disorders, applying regression and causal analysis to assess trends in Insomnia and Sleep Apnea prevalence

Data Engineer, LTIMindtree

Jan 2021 - Jun 2023

- Streamlined **policy data processing** by 40% through optimized **ELT pipelines** in **Azure Data Factory**, and Apache Airflow, enabling real-time access to 10M+ transactions for insurance underwriting teams
- Improved claims validation efficiency by 35% through optimized **Snowflake & Azure Synapse** transformations, reducing delays in insurance payouts
- Minimized policy reporting disruptions by 30% by automating end-to-end data workflows in **Apache Airflow** and **Python**, ensuring uninterrupted insights for business teams
- Accelerated dashboard response times by 40%, streamlining policy assessments for policy underwriters through optimized SQL queries and data transformations in MS SQL Server
- Optimized **cloud storage efficiency** by implementing data partitioning, compression, and indexing strategies in Snowflake, reducing computation costs by 20% while improving query performance

ACADEMIC PROJECTS

AI-Powered Resume Analysis Tool | *LangChain, OpenAI embeddings, Pinecone, Streamlit*

Dec 2024

- Automated resume analysis using **LangChain** and Pinecone as vector database, potentially easing out candidate screening process for recruiters by enabling intelligent search and matching
- Built an interactive **Streamlit dashboard**, allowing recruiters to query candidates based on job descriptions, potentially reducing **hiring turnaround time**

Statistical Analysis of Narrative Impact on Book Popularity | *OLS regression, KLD computation*

May 2024 – Jun 2024

- Analyzed narrative structure's effect on book popularity across 25,000 books, using **Kullback-Leibler Divergence (KLD)** and book-level measures like skewness, kurtosis, and sentiment volatility to assess reader engagement
- Identified key predictors of book downloads by implementing **OLS and LASSO regression models**, providing data-driven insights for authors and publishers on factors influencing reader preferences

Motor Vehicle Crash Data Integration and Analysis | *ETL Automation, SCD2, Talend*

Mar 2024- Apr 2024

- Enabled accident trend analysis by automating **ETL pipelines** in **Talend**, integrating 3M+ crash records into **Azure SQL** Server, ensuring structured reporting and potentially eliminating manual data aggregation and reducing crash occurrences
- Tracked historical accident trends by implementing **SCD2 dimensional model**, allowing yearly comparisons to identify high-risk zones and recurring crash patterns, supporting data-driven decision-making for traffic safety improvements